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Classification with Imbalanced Data using Python

For this task, I have used dataset related to Credit Risk. The main objective of this task was to build a classification model on an imbalance data using Python and libraries like **sklearn** and **imblearn**.

SMOTE (Synthetic Minority Over-sampling Technique) is a data resampling method in machine learning to address the problem of class imbalance. It creates minority class samples.

First , I have imported important libraries – **pandas , numpy , matplotlib , scikit-learn and imblearn.**

Then imported the dataset , after importing , have done basic necessary operations like checking first and last few rows for data consistency , datatypes check , summary checking and shape of dataset. Then , checked for missing and duplicated data. There were no duplicates found also , there were no missing values.

A bar chart was used to visualize the class distribution before **applying SMOTE**.

Categorical columns like employment type , education level, region , device type , were converted into numeric .

Then , **features and target** variable were separated. The dataset was divided into **training and testing** sets.80% of the data was used for training and 20% for testing.

The dataset had an imbalance between the classes. SMOTE was applied to the training data.

After applying **SMOTE**, another bar chart was created to visualize the **balanced class** distribution. The visualization confirmed that , the dataset became balanced.

A **classification model** was built using **Random Forest**. The model was trained using the balanced training dataset.

Model Evaluation :

The model was evaluated using the test dataset.
Confusion Matrix , Recall ,F1 Score were checked for model evaluation.